

Question Number	Answer	Additional Guidance	Mark
1(a)	An explanation that includes two of the following points: • polymer containing lots of glucose molecules (1) • a large molecule so that it {is insoluble / does not have an osmotic effect} (1) • branched molecule so that it can be hydrolysed quickly (1)	ALLOW polysaccharide ALLOW made of amylose and amylopectin as meaning large ALLOW broken down IGNORE easily	(2) exp

Question Number	Answer	Additional Guidance	Mark
1(b)	A description that includes five of the following points: • method to produce (same) initial {mass/volume/length} of cells (1) • Five different concentrations used/stated (1) • suitable control of temperature or a stated temperature (1) • measure change in mass /volume/length after the {same/stated} time interval (1) • repeats (at each concentration) and calculate mean (1) • plot change in mass/volume/length against concentration of solution/find point at which the (two)lines cross the x axis (1)	IGNORE size ALLOW air-conditioned room ALLOW time period in the range 30 mins to 3 days ALLOW calculation of mean size ALLOW size	(5) exp

Question Number	Answer	Additional Guidance	Mark
1(c)(i)	An answer that includes the following points: • (abiotic variable) pH / temperature / mass/volume at time zero (1) • (biotic variable) same age / same part of tuber (1)		(2) exp

Question Number	Answer	Additional Guidance	Mark
1(c)(ii)	An answer that includes the following points: • variable with suitable control method described (1) • results are not valid / description of expected effect on the dependent variable (1)	IGNORE accuracy/precision / reliability / repeatability ALLOW Description of how membranes affected	(2) exp

Question Number	Answer	Additional Guidance	Mark
1(d)	An answer that includes three of the following points: • some {molecules / solutes/ions} may be different (in each species) (1) • different concentrations of each {molecule / solutes/ions} (in each species) (1) • potato cells with different concentration of solution/cytoplasm affects water potential (1) • cell walls may have different structures (1)	ALLOW mineral ions / minerals ALLOW mineral ions / minerals ALLOW higher solute concentration will have lower water potential ALLOW wall pressure / turgor pressure / pressure potential	(3) exp

(Total for question 1 = 14 marks)